



Office of
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of Maritime Affairs

THE REPUBLIC OF LIBERIA

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Marine Advisory: 19/2025

Subject: High-Pressure Water Mist Nozzles – Corrosion Risk and Potential Failure

Dear Shipowner/Operator/Master/Recognized Organization:

The purpose of this Marine Advisory is to raise awareness regarding the [United States Coast Guard \(USCG\) Safety Alert 16-25](#), addressing potential failures in high-pressure water mist nozzles due to corrosion (green patina build-up), and to advise operators to ensure proper inspection and maintenance prior to Port State Control (PSC) inspections in U.S. ports

Requirements

The campaign applies to all ships and waterborne facilities that contain enclosed spaces. These spaces are defined as having one or more of the following characteristics:

- Limited entry and exit points
- Insufficient ventilation
- Not intended for continuous human occupation

Background

During a routine inspection of a fixed firefighting system aboard a foreign passenger vessel, USCG Port State Control Officers identified multiple “HI-FOG” system high-pressure water mist nozzles (Marioff “a2000” series) that failed to discharge water. The system failed to activate in several accommodation spaces.

The investigation revealed that corrosion—evident by green patina build-up had rendered the nozzles inoperable, even after the glass bulb actuator had broken as intended. The nozzle design relies on holding a standing pressure of 25 bar (363 PSI) and activating further to 140 bar (2,031 PSI) for full mist discharge. However, even after manually increasing the system pressure, as per manufacturer guidelines, the affected nozzles failed to release water.

The manufacturer, Marioff, previously issued Service Bulletin 001/2012 and Quality Bulletin 002/2012 related to similar failures in the “a1000” series. The current incident suggests these issues also apply to the “a2000” series.

Recommendations

The US Coast Guard recommends the following actions for vessel owners, operators, technical managers, and crew:

- Inspect all high-pressure water mist nozzles on board for corrosion or green patina—particularly those over 10 years old or located in high-moisture environments (e.g., outer decks, galleys, spas, pool areas, and near exterior doors).
- Review crew training procedures to ensure personnel responsible for firefighting equipment are able to identify signs of nozzle corrosion during routine maintenance or walk-throughs.
- Conduct additional inspections, if necessary, beyond scheduled maintenance intervals—particularly in high-risk areas.
- Consult the manufacturer (Marioff) for guidance if corrosion or malfunction is found. Replace nozzles as needed.
- Be aware that increasing system pressure to 140 bars may not always restore nozzle functionality, even when recommended by the manufacturer.

- This Safety Alert is intended for awareness and does not supersede any statutory or regulatory requirements. For full details, refer to USCG Safety Alert 16-25, and Marioff Service/Quality Bulletins 001/2012 and 002/2012 ([U.S. Coast Guard Safety Alert 16-25: Lookout for Green Patina Build-Up on High Pressure Water Mist Nozzles](#))

If you have any questions, please contact our Fleet Performance Department by Telephone: +1-703-790-3434 or by email to prevention@liscr.com.